THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The four angles of a quadrilateral are in AP with common difference 50°. The smallest angle is: 1

(a) 15° **(b)** 35° **(c)** 65° **(d)** 25°

2. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 25 cm. Its perimeter is: 1

(a) 95 cm **(b)** 105 cm **(c)** 75 cm **(d)** 70 cm

3. The diagonal of a square is $18\sqrt{2}$ cm. Its perimeter is: 1

(a) 90 cm **(b)** 80 cm **(c)** 72 cm **(d)** 108 cm

4. A rhombus has side 51 cm and one diagonal 48 cm. The other diagonal is: 1

(a) 90 cm **(b)** 135 cm **(c)** 92 cm **(d)** 88 cm

5. The diagonals of a rhombus are 28 cm and 96 cm. Its area is: 1

(a) 1372 cm^2 **(b)** 1354 cm^2 **(c)** 1344 cm^2 **(d)** 1440 cm^2

6. The diagonal of a rectangle is 25 cm and one side is 7 cm. Its area is: 1

(a) 175 cm^2 **(b)** 168 cm^2 **(c)** 192 cm^2 **(d)** 178 cm^2

7. In a parallelogram, two opposite angles are $(3x + 14)^\circ$ and $(x + 188)^\circ$. The value of x is: 1

(a) 89 **(b)** 86 **(c)** 88 **(d)** 87

8. The diagonals of a quadrilateral are 54 cm and 77 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 131 cm **(b)** 264 cm **(c)** 393 cm **(d)** 196 cm

9. **Assertion (A):** The sum of the angles of a quadrilateral is 360°. 1

Reason (R): A diagonal divides a quadrilateral into two triangles, each with angle sum 180°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The diagonals of a parallelogram bisect each other. 1

Reason (R): The diagonals of a parallelogram are always equal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A rhombus has side 51 cm and one diagonal 48 cm. Find the other diagonal, its area and the distance between a pair of opposite sides (height), correct to two decimal places. 2

12. The angles of a quadrilateral are in AP with common difference 40°. Find all four angles and say whether a pair of them is supplementary. 2

13. In parallelogram ABCD, $AB = (2x + 1)$ cm, $CD = (x + 23)$ cm and $BC = (x + 2)$ cm. Find x, the sides and the perimeter. 2

14. A rhombus has side 25 cm and one diagonal 14 cm. Find the other diagonal, its area and the distance between a pair of opposite sides (height), correct to two decimal places. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The diagonals of a parallelogram ABCD meet at O. $OA = (5x + 3)$ cm and $OC = (x + 147)$ cm. Find x and the length of AC. 3

16. In rhombus ABCD, $\angle A = 88°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

17. In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 36$ cm and $AB = 36$ cm. Find DE, AD and show that E is the midpoint of AC. 3

OR

The diagonals of a parallelogram ABCD meet at O. $OA = (4x + 3)$ cm and $OC = (2x + 57)$ cm. Find x and the length of AC. 3

18. In rhombus ABCD, $\angle A = 142°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

OR

In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 18$ cm and $AB = 32$ cm. Find DE, AD and show that E is the midpoint of AC. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A land surveyor in Nagpur marks a plot ABCD whose four angles are in the ratio $2 : 3 : 4 : 6$. 5

(i) What is the sum of the four angles?
(ii) Find the value of one part of the ratio.
(iii) Find the largest angle.
(iv) Is the plot a parallelogram? Give a reason.

20. The midpoints of the sides of a rectangle 42 cm by 144 cm are joined in order. Show that the figure formed is a rhombus and find its perimeter and area. 5

OR

In a parallelogram ABCD the diagonals meet at O with $OA = 3$ cm, $OB = 4$ cm and $AB = 5$ cm. Show that ABCD is a rhombus and find its perimeter. 5

END OF QUESTION PAPER — **Total: 40 Marks**